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Bayesian Optimization of control parameters for a 7 d.o.f. manipulator robot

Technical report
Machine Learning for Mechanical Systems

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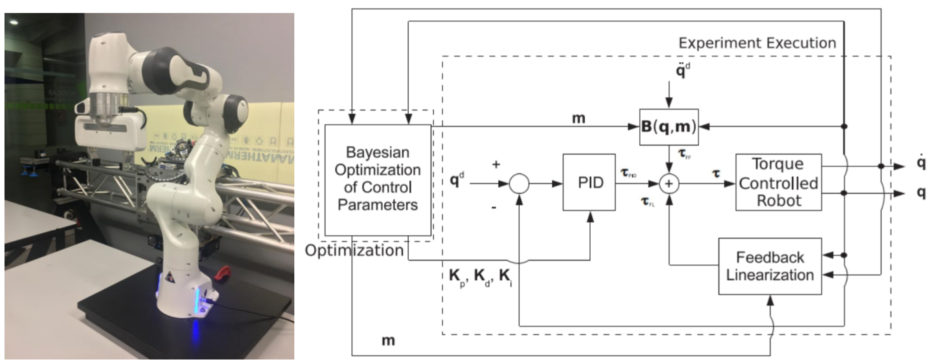
1 | Introduction

This report presents the implementation of a Bayesian Optimization (BO) framework for the tuning of control parameters in a 7-joint industrial manipulator. The objective of the present work is to design and evaluate a custom cost function tailored to accurately reflect specific performance criteria.



(a) (b)

Figure 1: 7 Dofs manipulator (a), control scheme (b)

The study is based on a pre-existing simulation environment, accompanied by reference code and datasets. These resources enable a focused investigation on the design of the optimization strategy, with particular attention to parameter selection and algorithmic efficiency.

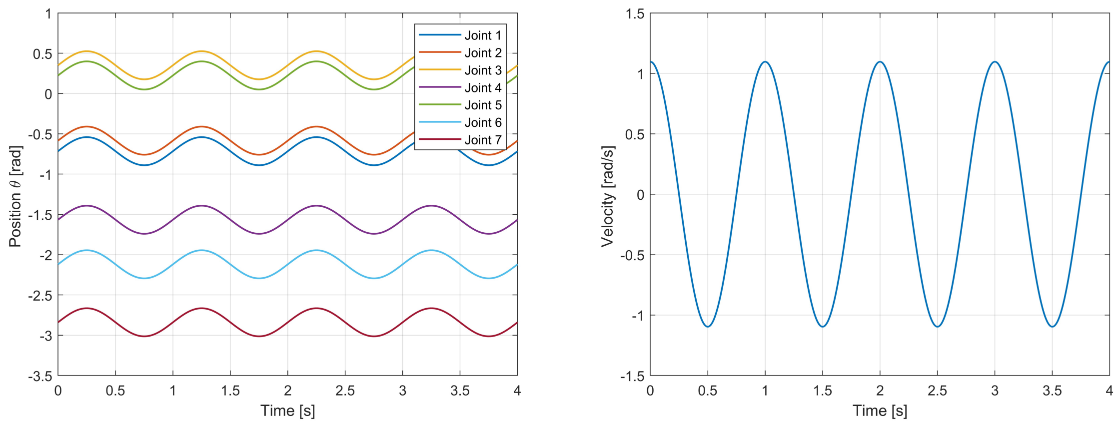
The overall procedure is structured into three main phases:

- **Cost Function Design** : A custom performance index is formulated and implemented, aiming to balance accuracy, stability, and trajectory smoothness, while incorporating penalty terms to mitigate instability or divergence. Two cost functions are defined to reflect different control perspectives: one operating in the joint space, focusing on individual joint behavior, and the other in the workspace, prioritizing end-effector accuracy. The specific goal in this stage is to determine the set of PID parameters for each joint motor.

- **Sequential Joint Optimization** : Rather than performing a joint optimization of all 21 PID parameters, a sequential approach is adopted. Each joint is optimized individually, starting from the end-effector and proceeding backward along the kinematic chain, with the goal of reducing dimensionality and improving the overall optimization efficiency.
- **Mass Parameter Estimation** : A Bayesian optimization process is employed to estimate the link masses based on acceleration tracking and torque data, carried out under an open-loop control condition. Accurate mass estimation enables improved model-based compensation of gravitational and inertial effects. After the masses are identified, a second round of PID tuning is performed, analogous to the second phase, to exploit the enhanced model fidelity and improve control performance.

2 | Joint-space Cost Function

Initially, the reference kinematics is generated using periodic motion laws for each degree of freedom (see fig. 2).



(a) (b)

Figure 2: Target cinematic reference a) trajectory reference profile for each joint, b) velocity reference profile shared between each joint

At each iteration, the code simulates the dynamics based on the current PID values and subsequently computes the cost function by evaluating different types of deviations between the reference and the simulated dynamics. In this case, the masses of the robot were assumed to be known. The first objective function used in this study, denoted as J, quantifies the tracking performance and control quality of the 7-DoF manipulator under a PID-based control architecture. It is defined as a weighted sum of normalized performance terms, capturing tracking errors, control smoothness, and actuation effort, with additional penalties to discourage instability.

The complete cost function is given by:

$$J = \mathbf{w}^{\top} \cdot \sum_{i=1}^{7} \boldsymbol{\varepsilon}^{i} + P_{max} + P_{prop} \tag{1}$$

where

$$\mathbf{w} = \begin{bmatrix} w_{e,\text{max}} \\ w_{\dot{e},\text{max}} \\ w_{e,\text{avg}} \\ w_{\dot{e},\text{avg}} \\ w_{\text{jerk,avg}} \\ w_{\tau,\text{max}} \end{bmatrix}, \quad \boldsymbol{\varepsilon}^i = \begin{bmatrix} e^i_{\text{max}} \\ \dot{e}^i_{\text{max}} \\ e^i_{\text{avg}} \\ \dot{e}^i_{\text{avg}} \\ e^i_{\text{jerk,avg}} \\ \tau^i_{\text{max}} \end{bmatrix} \tag{2}$$

The vector $\mathbf{w}$ contains the weights associated with each error term, they are coefficients used to balance the importance of each contribution in the optimization process. The weight set was determined based on general principles and then slightly adjusted through a trial-and-error process (see tab. 1). In general, higher weights were assigned to position-related errors to ensure accurate tracking. These are followed, in decreasing order of magnitude, by the weights associated with velocity errors and then by the jerk term, which, despite its lower weight, remains important to ensure motion smoothness. The lowest weight is assigned to the torque-related term, as it is generally desirable to limit the structural stress on the actuators and reduce energy consumption.

$w_{e,\text{max}}$	$w_{\dot{e},\text{max}}$	$w_{e,\text{avg}}$	$w_{\dot{e},\text{avg}}$	$w_{\text{jerk,mean}}$	$w_{\tau,\text{max}}$
50	1	30	5	1	0.02

Table 1: Weight coefficients used in the joint-cost function.

Moreover, the terms of vector ε^i are here defined, note each error is normalized in order to give the same magnitude to each error component:

- $e^i_{\text{max}} = \max_t \left| \frac{q_{\text{ref},i}(t) - q_i(t)}{q_{r,\text{amp}}} \right|$: maximum normalized i-th joint position errors.
- $\dot{e}^i_{\text{max}} = \max_t \left| \frac{\dot{q}_{\text{ref},i}(t) - \dot{q}_i(t)}{\dot{q}_{r,\text{amp}}} \right|$: maximum normalized i-th joint velocity errors.
- $e^i_{\text{avg}} = \frac{1}{T} \int_0^T \left| \frac{q_{\text{ref},i}(t) - q_i(t)}{q_{r,\text{amp}}} \right| dt$: average normalized i-th joint position error.
- $\dot{e}^i_{\text{avg}} = \frac{1}{T} \int_0^T \left| \frac{\dot{q}_{\text{ref},i}(t) - \dot{q}_i(t)}{\dot{q}_{r,\text{amp}}} \right| dt$: average normalized i-th joint velocity error.
- $e^i_{\text{jerk,avg}} = \frac{1}{T} \int_0^T \left| \frac{\dddot{q}_{ref,i}(t) - \dddot{q}_i(t)}{\dddot{q}_{r,\text{amp}}} \right| dt$: average normalized i-th joint jerk error.
- $\tau^i_{\text{max}} = \max_t \left| \frac{\tau_{\text{PID},i}(t) + \tau_{\text{comp},i}(t)}{\tau_{\text{max},i}} \right|$: maximum normalized torque applied.

The last two terms are penalties which are triggered when certain error exceed a maximum value assigned:

- P_{max} is a fixed term (set as 2000) assigned when the simulation is stopped due to error position greater than 15° or near-singular Jacobian condition number, e.g. when two consecutive joints are close to be perfectly aligned.
- P_p is a proportional penalty applied when one or more joints exhibit intermediate position errors between a lower (5°) and upper (15°) tolerance threshold. It is defined such that at lower limit the proportional error has zero value and at upper limit it is equal to P_{max}

The cost function also includes an *anti-windup* mechanism for the integral action of the PID controller: the integral term is updated only when the computed torque does not exceed the saturation limits. This ensures the absence of undesired accumulation of control effort when actuators are already saturated, avoiding overshoot and instability.

The variables $q_{r,\text{amp}}$, $\dot{q}_{r,\text{amp}}$, and $\dddot{q}_{r,\text{amp}}$ denote the amplitude of position, velocity, and jerk reference, respectively. The torque normalization factor $\tau_{\max,i}$ depends on the actuator limits for each joint.

This cost function enables a multi-objective tuning that accounts for tracking precision, control effort, motion smoothness, and system robustness. It is evaluated at each iteration of the Bayesian Optimization process to guide the selection of optimal PID parameters.

Once the cost function was defined, the original code was modified by implementing two separate BO phases: an initial one favoring exploration and a subsequent one favoring exploitation. Between the two simulations, the search space for the proportional gains K_p^i is narrowed based on the minimum value observed during the initial exploration phase. This adjustment is motivated by the empirical observation that the optimal K_p^i values consistently fall within a relatively high range, and are never close to zero. Therefore, excluding very low values helps focus the search in more promising regions and improves optimization efficiency.

The hyperparameters of the process are as follows:

- N° of initial seed points: 15
- N° of iterations in exploration phase: 150
- Acquisition function in exploration phase: `'lower-confidence-bound'`
- N° of iterations in exploitation phase: 50
- Acquisition function in exploitation phase: `'expected-improvement'`
- Processor parallelization: `''UseParallel', true'`

To assess the quality of the obtained results, the following set of metrics is employed:

- `mean_pos_abs` [deg]: Mean value of the absolute position error throughout the trajectory
- `max_pos` [deg]: Maximum value of the absolute position error reached during the motion
- `var_pos_abs` [deg]: Variance of the absolute position error
- `mean_vel_abs` [deg/s]: Mean value of the absolute velocity error throughout the trajectory
- `max_vel` [deg/s]: Maximum value of the absolute velocity error reached during the motion
- `var_vel_abs` [deg/s]: Variance of the absolute velocity error

2.1. Results

The results of the optimization process, based on the previously defined cost function, are now presented.

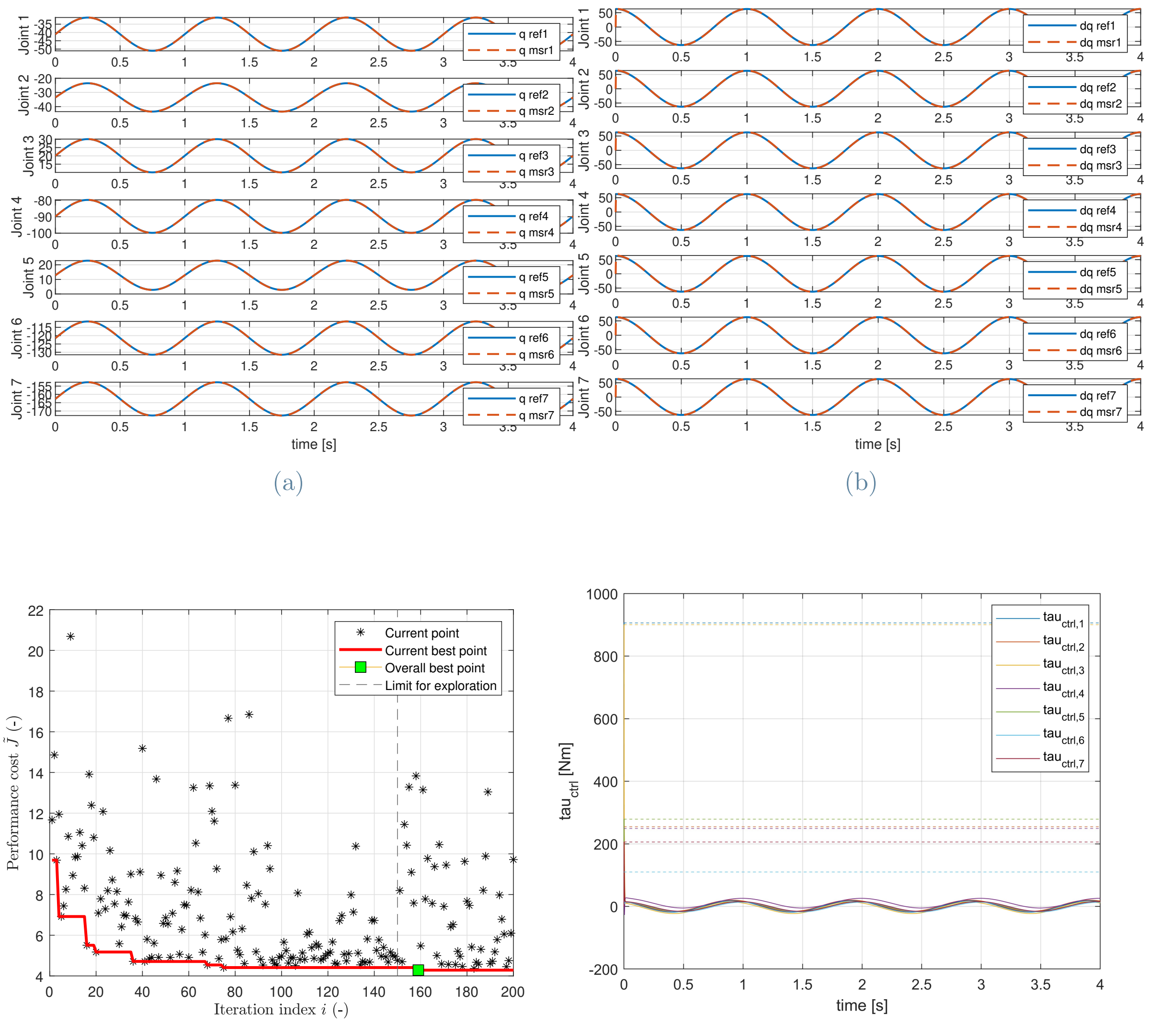


Figure 3: BO results a) trajectory profile for each joint compared against reference b) velocity profile compared against reference, c) Cost function vs iteration d) Joint torques profile

As shown in fig. 3a and fig. 3b, the algorithm is fully capable of tracking the generated reference trajectories. Optimal PID gains are shown in tab. 6, while the metrics for this optimization results are shown in tab. 2.

KPI	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7	Mean
mean_pos_abs [deg]	$4.7 \cdot 10^{-4}$	$5.9 \cdot 10^{-4}$	$5.2 \cdot 10^{-4}$	$6.1 \cdot 10^{-4}$	$6.5 \cdot 10^{-4}$	0.0004	$6.2 \cdot 10^{-4}$	0.00052
max_pos [deg]	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
var_pos_abs [deg]	$5.9 \cdot 10^{-8}$	$8.9 \cdot 10^{-8}$	$7.0 \cdot 10^{-8}$	$9.5 \cdot 10^{-8}$	$1.0 \cdot 10^{-7}$	$4.9 \cdot 10^{-8}$	$9.6 \cdot 10^{-8}$	$8.1 \cdot 10^{-8}$
mean_vel_abs [deg/s]	0.0039	0.0038	0.0042	0.0042	0.0040	0.0040	0.0040	0.0040
max_vel [deg/s]	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
var_vel_abs [deg/s]	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004

Table 2: Summary of selected KPI metrics baseline case

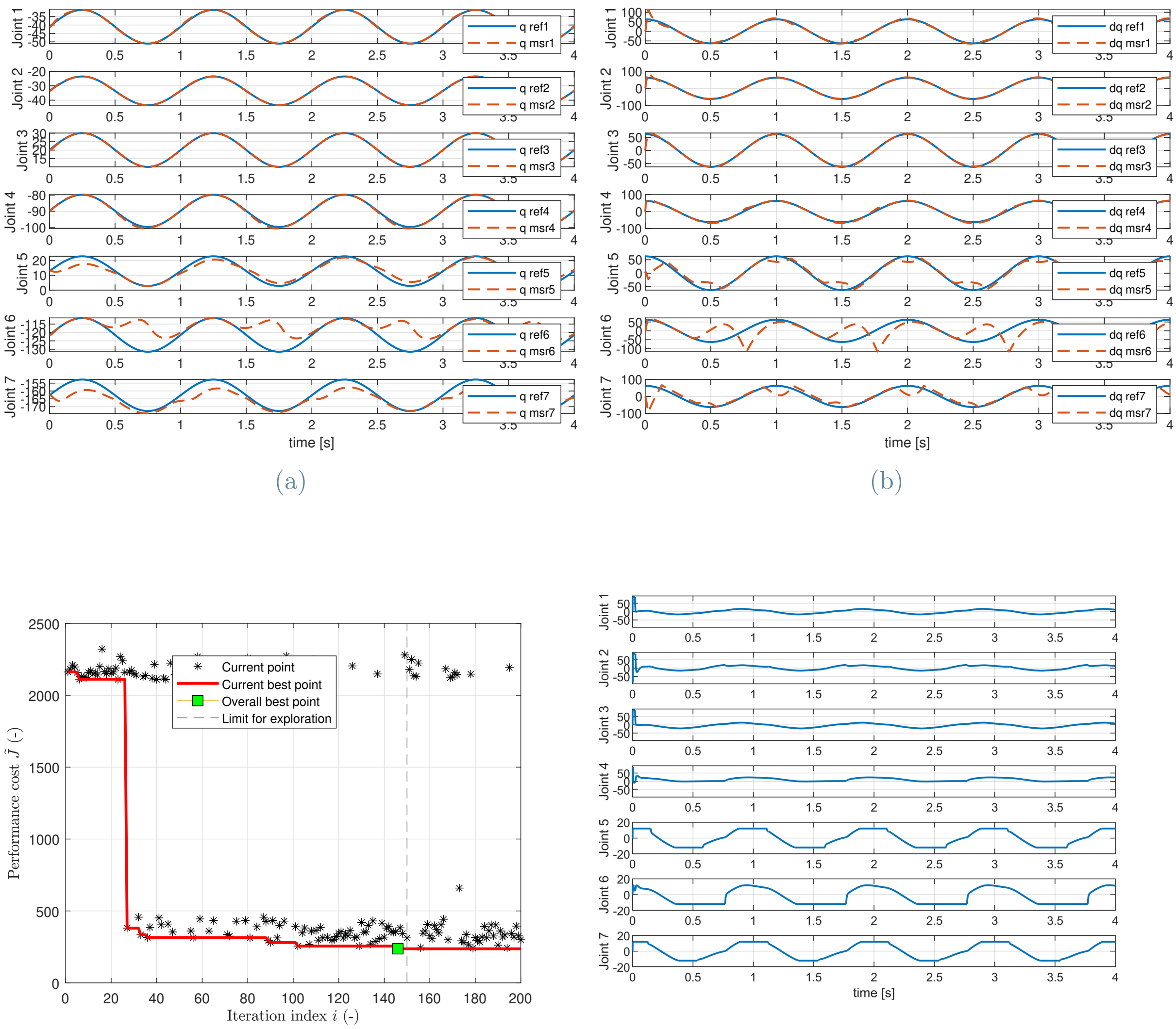
It can be observed from fig. 3d that the torques applied by the motors exceed the nominal limit values (tab. 3). To increase realism, actuator saturation was introduced by limiting the maximum torque for each joint, based on real robot's physical specifications.

	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7
Max torque	87 Nm	87 Nm	87 Nm	87 Nm	12 Nm	12 Nm	12 Nm

Table 3: Maximum torque values for each joint

2.2. Active Saturation Results

The system is now constrained to apply, at most, the specified torque values, maintaining the maximum value in cases where the required torque exceeds the limit.



(a) (b)

(c) (d)

Figure 4: Optimization results with active saturation: a) trajectory profile for each joint compared against reference b) velocity profile compared against reference, c) Cost function vs iteration d) Joint torques profiles

As expected, the results are slightly worse than in the first case, with the new optimal PID gains shown in tab. 7 and metrics shown in tab. 4.

KPI	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7	Mean
mean_pos_abs [deg]	0.0028	0.0015	0.0010	0.0065	0.034	0.089	0.052	0.0256
max_pos [deg]	0.017	0.0078	0.0095	0.032	0.11	0.32	0.16	0.0985
var_pos_abs [deg]	$1.2{\cdot}10^{-5}$	$2.4{\cdot}10^{-6}$	$1.8{\cdot}10^{-6}$	$5.1{\cdot}10^{-5}$	$6.8{\cdot}10^{-4}$	$1.1{\cdot}10^{-2}$	$2.0{\cdot}10^{-3}$	$1.67 \cdot 10^{-3}$
mean_vel_abs [deg/s]	0.042	0.022	0.016	0.054	0.19	0.64	0.28	0.226
max_vel [deg/s]	1.1	1.1	1.1	1.5	1.5	2.2	2.1	1.41
var_vel_abs [deg/s]	0.0060	0.0029	0.0020	0.0038	0.039	0.11	0.092	0.0355

Table 4: KPI values for active saturation

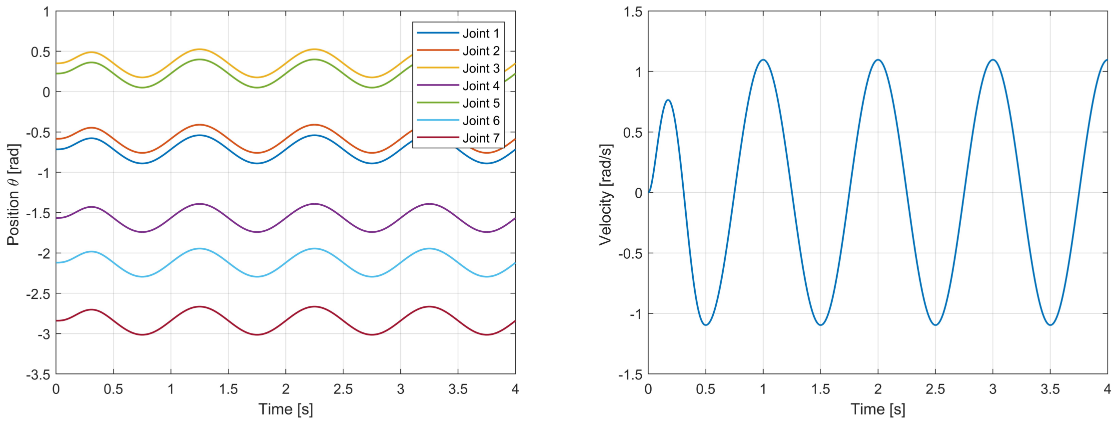
In particular, it can be observed that while the first four joints show results similar to the baseline case, the last three joints (which are subject to a torque limit of 12 Nm) are unable to accurately follow the target references. This occurs because the torques required to achieve those references exceed the imposed limit, resulting in errors in terms of motion amplitude and delays in the response.

A remark should be made about the velocity profiles: fig. 4b illustrates that for the first four joints, at the beginning of the motion, an initial overshooting in velocity occurs. However, this overshoot is quickly corrected, and the velocity then closely follows the reference. Also the last three joints present a similar error in initial motion, but however they aren't able to follow the reference due to limitations on the torque to be applied. This initial phenomenon is due to the fact that the reference imposes a non-zero initial value for velocity (see fig. 2b), which constitutes a physical inconsistency.

2.3. Adjusted Reference Results

To further approximate real-world behavior, the reference trajectory was modified to ensure physically plausible profiles (fig. 5a). Specifically, in order to obtain a velocity profile that starts from zero, a segment of the position profile (from the starting time up to $t = 0.5\,\mathrm{s}$) was replaced by a 6th-degree polynomial satisfying the following conditions:

- At $t = 0$: position, first and second derivatives set to zero;
- At $t = 0.5\,\mathrm{s}$: continuity is imposed on position and on the first, second and third derivatives.



(a) (b)

Figure 5: New modified kinematic reference: a) trajectory reference profile for each joint, b) velocity reference profile shared between each joint

Upon re-running the optimization, the results displayed in fig. 6 are obtained.

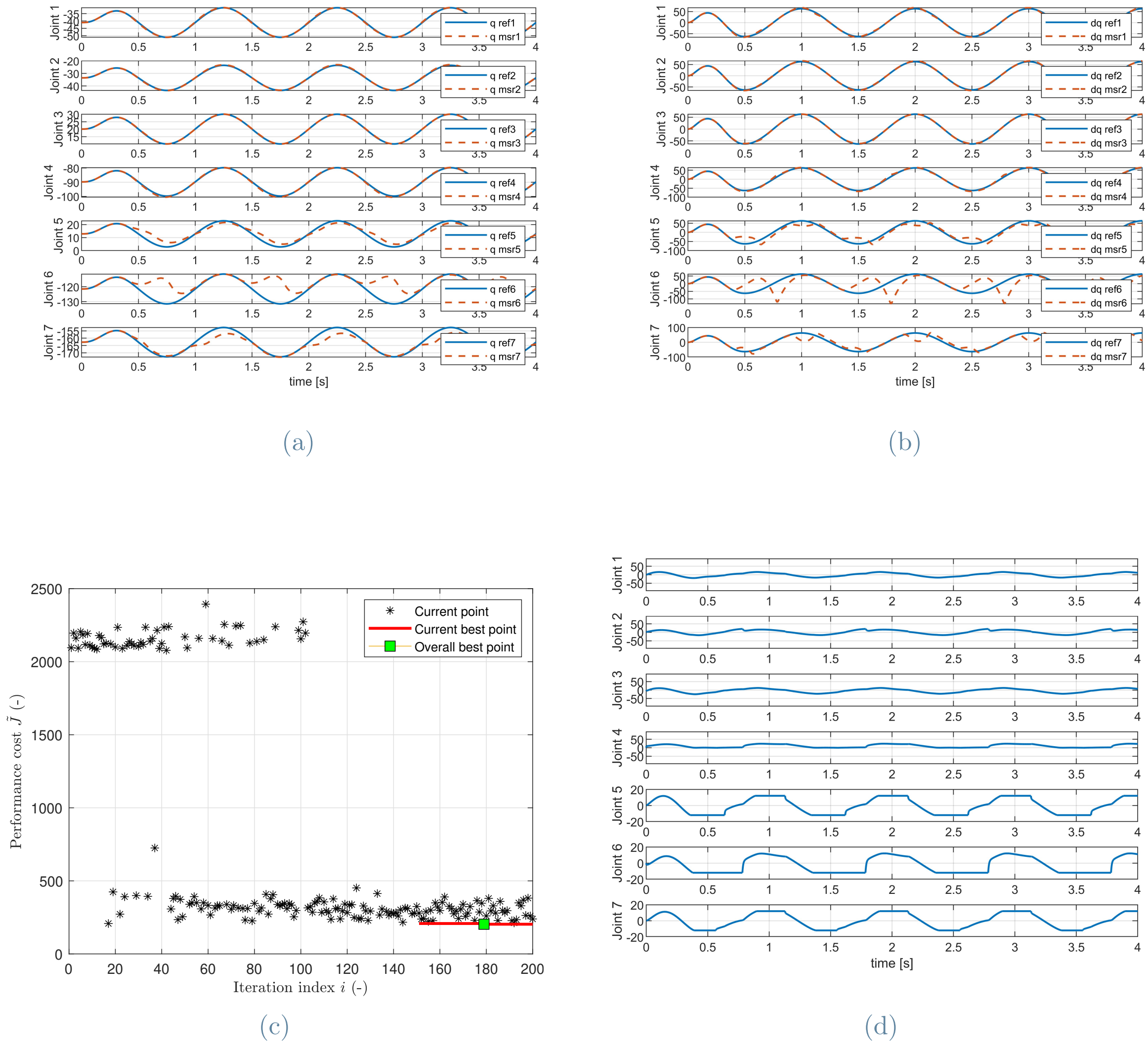


Figure 6: Optimization results with active saturation: a) trajectory profile for each joint compared against reference b) velocity profile compared against reference, c) Cost function vs iterations d) Joint torque profiles

The results continue to be worse than in the baseline case (optimal PID gains are shown in tab. 8), and metrics are shown in tab. 5..

KPI	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7	Mean
mean_pos_abs [deg]	0.0026	0.0037	$8.66 \cdot 10^{-4}$	0.0053	0.0333	0.0831	0.0306	0.0224
max_pos [deg]	0.0095	0.0100	0.0050	0.0180	0.1126	0.3218	0.1208	0.0854
var_pos_abs [deg]	$8.36 \cdot 10^{-6}$	$8.52 \cdot 10^{-6}$	$1.38 \cdot 10^{-6}$	0.0031	$6.49 \cdot 10^{-4}$	0.0109	0.0012	0.0029
mean_vel_abs [deg/s]	0.0311	0.0198	0.0114	0.0435	0.2460	0.6272	0.2302	0.1728
max_vel [deg/s]	0.1091	0.0666	0.0708	0.2076	0.6783	2.5564	0.9699	0.6655
var_vel_abs [deg/s]	$7.75 \cdot 10^{-4}$	0.0003	0.0014	0.0017	0.0288	0.4317	0.0609	0.0751

Table 5: KPI values for each joint for adjusted reference

However, modifying the reference in the initial phase resolves the initial mismatch both in the trajectory (fig. 4a vs fig. 6a) and in the velocity profiles (fig. 4b vs fig. 6b). This highlights that such error can be recovered by enforcing a more realistic kinematic profile. On the other hand, the issues observed in the remaining part of the motion are still caused by the motors reaching their saturation limits. In theory, this problem could be solved by increasing the maximum torque deliverable by the motors of the last three joints, which would allow for perfect coherence, as seen in the initial baseline case.

Here is a summary of the optimal set of PID parameter for each of the three cases presented so far.

	Joint 1	**Joint 2**	**Joint 3**	**Joint 4**	**Joint 5**	**Joint 6**	**Joint 7**
K_p	2122.1	4351.8	3348.2	3972.6	8549.1	817.1	6183.3
K_i	1070.0	1035.8	528.1	937.4	311.5	784.8	58.5
K_d	508.5	497.9	527.3	466.3	494.9	497.3	494.2

Table 6: Optimal PID parameter set of baseline case

	Joint 1	**Joint 2**	**Joint 3**	**Joint 4**	**Joint 5**	**Joint 6**	**Joint 7**
K_p	8896.9	9243.2	7888.2	8119.8	480.8	2095.9	970.6
K_i	367.1	805.9	682.4	71.4	120.2	537.9	300.8
K_d	351.8	248.5	388.5	418.9	317.4	238.1	477.1

Table 7: Optimal PID parameter set with active saturation

	Joint 1	**Joint 2**	**Joint 3**	**Joint 4**	**Joint 5**	**Joint 6**	**Joint 7**
K_p	9437.2	847.5	9515.2	8411.8	1132.6	1894.6	1182.2
K_i	374.1	27.2	684.0	828.9	107.7	473.4	730.5
K_d	459.0	508.9	402.0	305.4	287.3	171.7	375.9

Table 8: Optimal PID parameter set with active saturation and adjusted kinematic

3 | Work-space Cost Function

A second cost function was designed, not as a refinement of the first, but rather as an alternative formulation tailored to potential real-world requirements. In practical scenarios, such as pick-and-place robotics, the most critical performance criteria typically concern the motion accuracy of the tool center point, i.e., the end-effector, while joint-level behavior is of lesser importance.

To reflect this, the second cost function shifts the focus from joint-level quantities to task-space errors directly expressed in cartesian coordinates. Specifically, the formulation penalizes the position, velocity, and jerk of the end-effector, thereby prioritizing tracking accuracy at the TCP. This approach aligns more closely with operational requirements, where the effectiveness of the robot is ultimately judged by the precision of its interaction with the environment, rather than by the internal dynamics of its joints.

The cost function to be minimized is given by:

$$J = \mathbf{w}_2^\top \cdot \sum_{i=1}^{3} \boldsymbol{\varepsilon}_2^i + P_{\max} + P_{prop} \tag{3}$$

where:

$$\mathbf{w_2} = \begin{bmatrix} w_{e,\max} \\ w_{\dot{e},\max} \\ w_{e,\text{avg}} \\ w_{\dot{e},\text{avg}} \\ w_{\text{jerk,avg}} \\ w_{\tau,\max} \end{bmatrix}, \quad \boldsymbol{\varepsilon}_2^i = \begin{bmatrix} e^i_{\max} \\ \dot{e}^i_{\max} \\ e^i_{\text{avg}} \\ \dot{e}^i_{\text{avg}} \\ e^i_{\text{jerk,avg}} \\ \tau^i_{\max} \end{bmatrix}$$

Each term in the vector $\boldsymbol{\varepsilon}_2^i$ contributes as follows:

- $e^i_{\max} = \max \left| \frac{s_{r,i}(t) - s_{msr,i}(t)}{s_{r,i,\text{amp}}} \right|$: maximum normalized cartesian position error.
- $\dot{e}^i_{\max} = \max \left| \frac{\dot{s}_{r,i}(t) - \dot{s}_{msr,i}(t)}{\dot{s}_{r,i,\text{amp}}} \right|$: maximum normalized cartesian velocity error.
- $e^i_{\text{avg}} = \frac{1}{T} \int \left| \frac{s_{r,i}(t) - s_{msr,i}(t)}{s_{r,i,\text{amp}}} \right| dt$: average normalized cartesian position error.

- $\dot{e}^i_{\text{avg}} = \frac{1}{T}\int \left|\frac{\dot{\mathbf{s}}_{r,i}(t)-\dot{\mathbf{s}}_{msr,i}(t)}{\dot{\mathbf{s}}_{r,i,\text{amp}}}\right| dt$: average normalized cartesian velocity tracking error.
- $e^i_{\text{jerk,avg}} = \frac{1}{T}\int_0^T \left|\frac{\dddot{s}_{,i}(t)-\dddot{s}_{msr,i}(t)}{\dddot{s}_{r,i,\text{amp}}}\right| dt$: average normalized jerk error.
- $\tau^i_{\text{max}} = \max_t \left|\frac{\tau_{\text{PID},i}(t)+\tau_{\text{comp},i}(t)}{\tau_{\text{max},i}}\right|$: maximum normalized torque applied.

The variables $s_{r,i,\text{amp}}$, $\dot{s}_{r,i,\text{amp}}$, and $\dddot{s}_{r,i,\text{amp}}$ denote the expected amplitude of position, velocity, and jerk signals, respectively along x,y and z cartesian coordinates. The torque normalization factor $\tau_{\text{max},i}$ depends on the actuator limits for each joint.

The vector of weights $\mathbf{w}_2$ (tab. 9) tunes the relative importance of each term. These weights are selected to prioritize positional accuracy, followed by velocity and smoothness, while keeping actuator efforts low.

$w_{e,\text{max}}$	$w_{\dot{e},\text{max}}$	$w_{e,\text{avg}}$	$w_{\dot{e},\text{avg}}$	$w_{\text{jerk,avg}}$	$w_{\tau,\text{max}}$
20	4	40	10	2	1

Table 9: Weight coefficients used in the work-space cost function

The last two terms are penalties which are triggered when certain error exceed a maximum value assigned:

- P_{max}: A large fixed penalty (e.g., 2000) added if the simulation is terminated early due to violation of a maximum cartesian error threshold or encountering a singularity.
- P_{p}: A proportional penalty added when the cartesian tracking error exceeds a soft lower threshold but not the hard upper bound. It is defined such that at lower limit the proportional error has zero value and at upper limit it is equal to P_{max}

3.1. Results

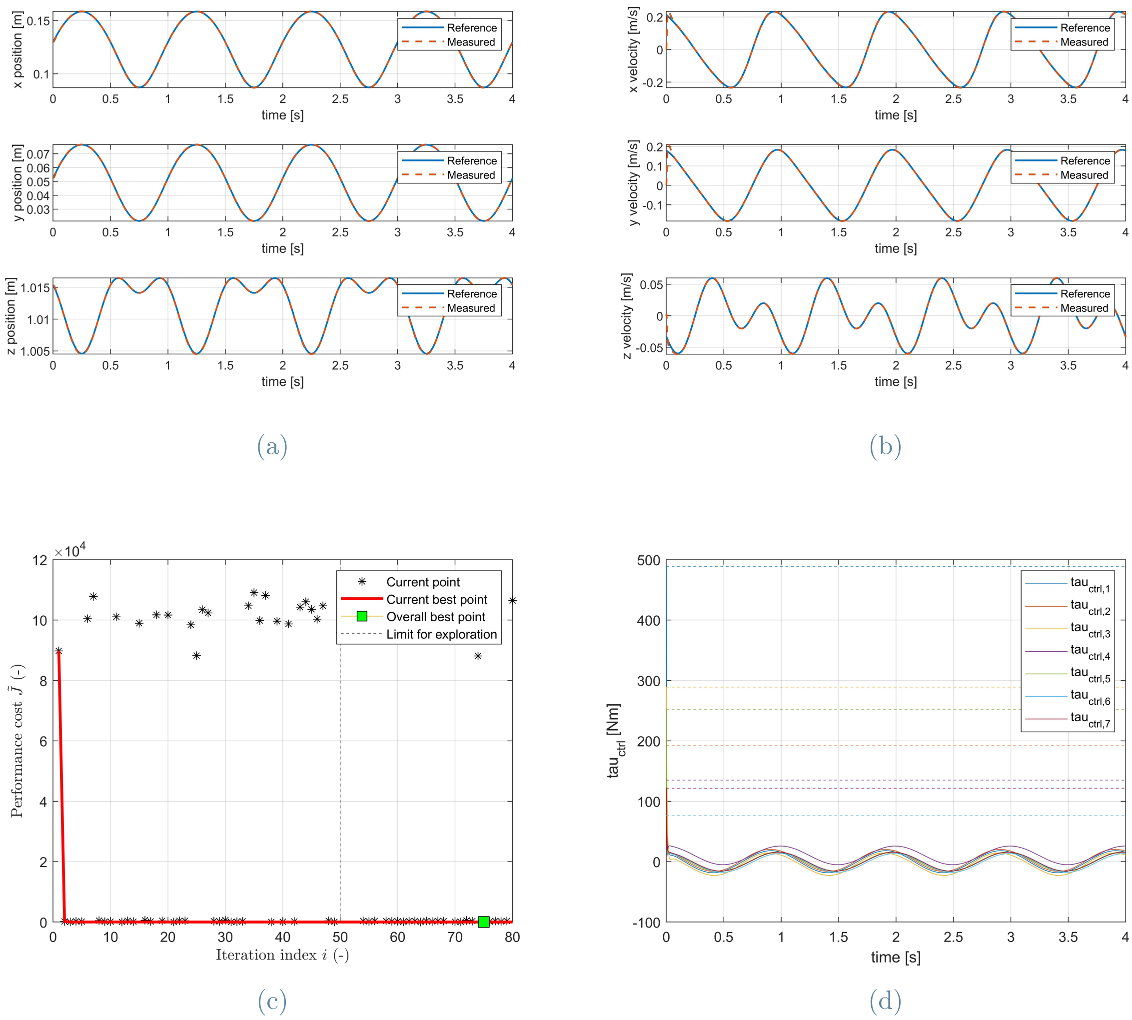


Figure 7: Optimization results for work-space cost function a) end-effector position tracking compared against reference b) end-effector velocity profile compared against reference, c) Cost function vs iterations d) Joint torque profiles

As can be observed from fig. 7a and fig. 7b, the previously defined workspace cost function is perfectly capable of accurately tracking the assigned position and velocity reference profiles. The same considerations made for the baseline case of the joint-space cost function regarding the applied torques remain valid for this type of optimization as well: the motors of each joint continue to require torques that exceed their nominal limits (see fig. 7d). The relative metrics and the optimized PID gains are shown respectively in tab. 10 and tab. 11

KPI	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7	Mean
mean_pos_abs	7.20e-4	7.65e-4	8.48e-4	7.14e-4	6.68e-4	7.15e-4	7.74e-4	7.45e-4
max_pos	0.0018	0.0023	0.0037	0.0018	0.0013	0.0018	0.0021	0.0021
var_pos_abs	1.28e-7	1.46e-7	1.97e-7	1.23e-7	1.09e-7	1.24e-7	1.50e-7	1.40e-7
mean_vel_abs	0.0048	0.0052	0.0064	0.0043	0.0040	0.0044	0.0055	0.0050
max_vel	1.10	1.10	1.10	1.10	1.10	1.10	1.10	1.10
var_vel_abs	5.17e-4	6.26e-4	9.41e-4	5.54e-4	4.43e-4	5.41e-4	5.71e-4	5.70e-4

Table 10: KPI values for each joint for work-space cost function

	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7
K_p	6000.1	6895.6	6718.8	9582.6	7033.5	8505.8	5177.9
K_i	110.12	738.08	673.6	117.12	125.38	123.41	482.52
K_d	348.56	275.31	172.62	320.08	428.16	329.49	307.63

Table 11: Optimal PID parameters set for work-space cost function

4 | Cascade Bayesian Optimization

4.1. Methodological Approach

The script implements a progressive optimization framework for tuning the PID controller parameters. The central idea is to decouple the high-dimensional optimization problem by tuning one joint at a time, thus significantly reducing the complexity of the variable space and improving the optimal search efficiency.

At each iteration, starting from the end-effector joint, the optimization focuses on a single joint, while the previously optimized ones retain their best-found parameters. The remaining unoptimized joints are assigned nominal values (e.g., zero), effectively isolating the contribution of the current DOF to the overall cost function. This modular approach ensures that the BO process concentrates its exploration on a lower dimensional subspace while gradually building the complete controller.

Although the optimization process proceeds under the simplifying assumption that each joint can be tuned independently, this is not strictly accurate from a physical standpoint. In reality, the dynamics of serial manipulators are inherently coupled: the motion of one joint influences and is influenced by the others due to shared inertial, Coriolis, and gravitational effects. However, treating each joint as dynamically decoupled becomes a valid approximation under specific conditions, namely, when the unoptimized joints are initialized with small, stable gains that do not introduce significant disturbances, and when the reference trajectories are sufficiently smooth. This strategy assumes that the optimization of each joint can be performed in isolation, with the residual coupling effects being absorbed or compensated by the local PID controller. While this introduces a degree of suboptimality, it enables a scalable and tractable optimization process.

The hyperparameters chosen for each one of the seven consecutive BO are defined as follows:

- N° of initial seed points: 4
- N° of iterations in exploration phase: 32

- Acquisition function in exploration phase: `'lower-confidence-bound'`
- N° of iterations in exploitation phase: 32
- Acquisition function in exploitation phase: `'expected-improvement'`

The cost function adopted for this optimization process is the one previously defined as joint-space cost function.

4.2. Results

Here are shown the results for the cascade BO.

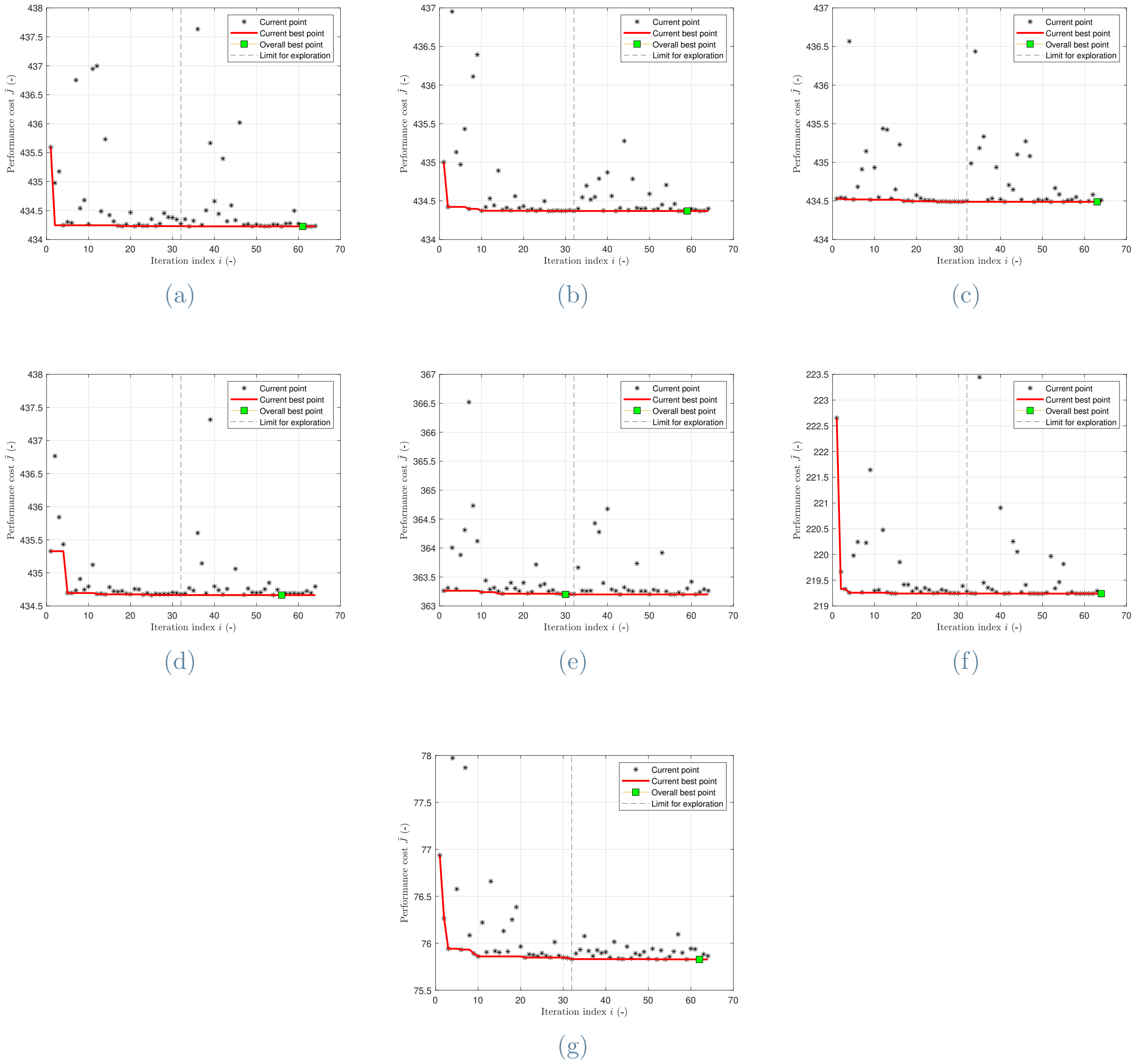
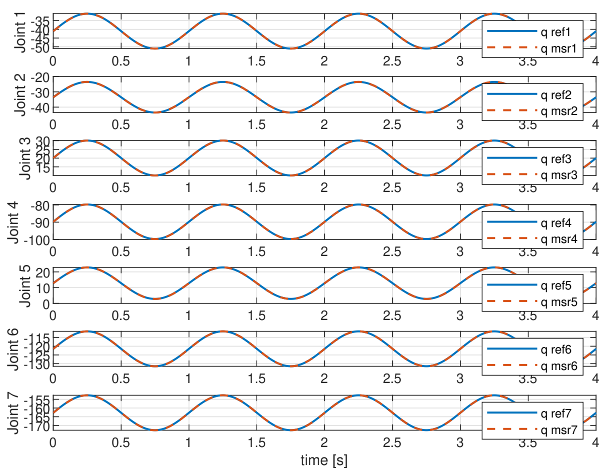
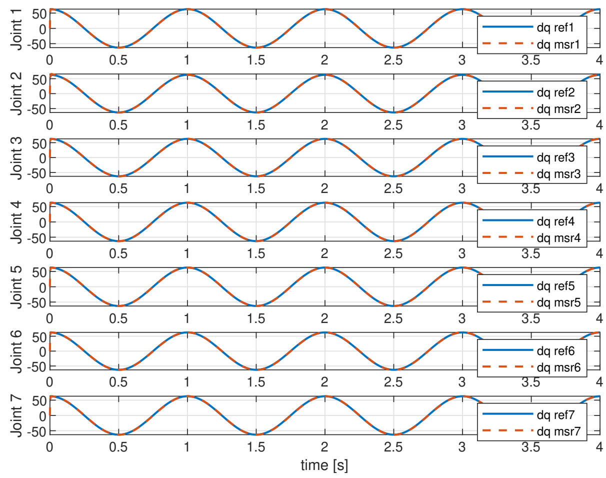


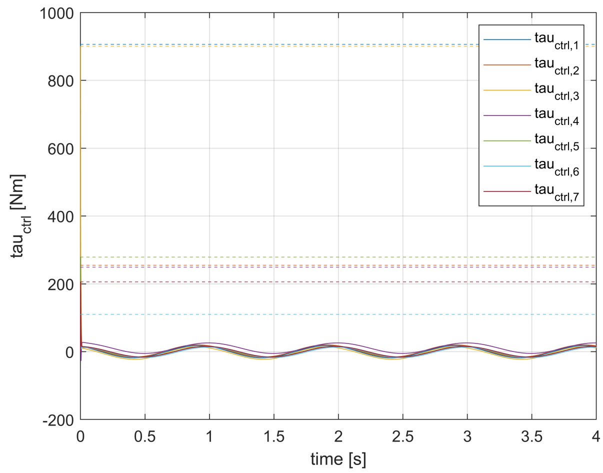
Figure 8: Cost function vs iterations for each joint in cascade PID tuning. a–g) Joints 1–7 optimized progressively.



(a)



(b)



(c)

Figure 9: Results from cascade optimization: a) trajectory profile compared against reference, b) velocity profile compared against reference, c) joint torque profiles

The optimal set of PID gains and the relative metrics are shown respectively in tab. 13 and in tab. 12.

As seen in fig. 9a and fig. 9b also the cascade is well suited for following the tracking reference. Further comment will be made in the conclusion chapter regarding the comparison between cascade approach and one-shot approach.

The optimal PID gains set and the relative metrics are shown respectively in tab. 13 and tab. 12

KPI	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7	Mean
mean_pos_abs	0.00028	0.00019	0.00028	0.0003	0.00038	0.00039	0.00039	0.00032
max_pos	0.00088	0.00080	0.00088	0.00088	0.0011	0.0011	0.0011	0.00099
var_pos_abs	$2.9 \cdot 10^{-8}$	$1.1 \cdot 10^{-7}$	$2.8 \cdot 10^{-8}$	$2.8 \cdot 10^{-8}$	$7.1 \cdot 10^{-8}$	$4.5 \cdot 10^{-8}$	$4.5 \cdot 10^{-8}$	$5.1 \cdot 10^{-8}$
mean_vel_abs	0.0041	0.0037	0.0041	0.0041	0.0049	0.0049	0.0049	0.0044
max_vel	1.097	1.097	1.097	1.097	1.097	1.097	1.097	1.097
var_vel_abs	0.00038	0.00045	0.00038	0.00038	0.00039	0.0004	0.0004	0.00039

Table 12: KPI values for each joint in cascade optimization configuration

	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7
K_p	735.81	10917	744.61	999.8	653.3	889.18	851.05
K_i	302.96	344.68	68.894	69.219	3.1029	31.674	57.719
K_d	549.72	422.42	549.99	548.08	499.98	499.94	499.97

Table 13: Optimal PID parameters set for cascade optimization

5 | Mass Estimation

5.1. Mass Parameter Estimation for Model-Based Compensation

In many practical scenarios, the exact values of the link masses required for model-based control, especially feedback linearization, are not precisely known. Accurate knowledge of these parameters is essential to properly compensate for gravity and Coriolis effects in the robot dynamics. Incorrect mass values can lead to inaccurate feedforward terms, ultimately reducing the effectiveness of control actions and degrading trajectory tracking performance.

To overcome this limitation, we propose a two-stage optimization approach. In the first stage, we estimate the link masses by exploiting the dynamic model, imposing a control input of the following form:

$$\boldsymbol{\tau} = \mathbf{B}(\mathbf{q}, \mathbf{m})\ddot{\mathbf{q}}_d + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{m})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}, \mathbf{m}) \tag{4}$$

In this way, if the mass parameters $\mathbf{m}$ are correctly estimated, the resulting dynamics satisfy:

$$\mathbf{B}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{B}(\mathbf{q}, \mathbf{m})\ddot{\mathbf{q}}_d + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{m})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}, \mathbf{m}) \quad \Rightarrow \quad \ddot{\mathbf{q}} = \ddot{\mathbf{q}}_d \tag{5}$$

We define a cost function that penalizes the deviation between the actual acceleration $\ddot{\mathbf{q}}$ and the desired acceleration $\ddot{\mathbf{q}}_d$. This cost is minimized with respect to $\mathbf{m}$ using BO, allowing for a data-efficient exploration of the parameter space.

Mass parameter tuning improves the quality of dynamics compensation, enabling more accurate trajectory tracking and reducing the effort required in the second optimization stage, which focuses on PID gain tuning.

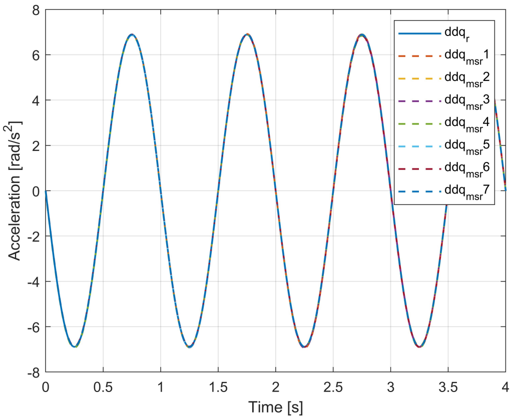
The hyperparameters of the mass estimation process are as follows:

- N° of initial seed points: 10
- N° of iterations in exploration phase: 100
- Acquisition function in exploration phase: `'lower-confidence-bound'`
- N° of iterations in exploitation phase: 100
- Acquisition function in exploitation phase: `'expected-improvement'`

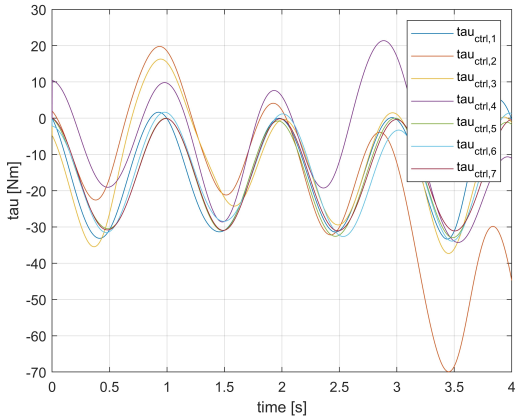
Once mass estimation is concluded, the following PID tuning is performed by exploiting the one-shot BO approach previously defined. The hyperparameters of the BO process are as follows:

- N° of initial seed points: 15
- N° of iterations in exploration phase: 150
- Acquisition function in exploration phase: `'lower-confidence-bound'`
- N° of iterations in exploitation phase: 50
- Acquisition function in exploitation phase: `'expected-improvement'`

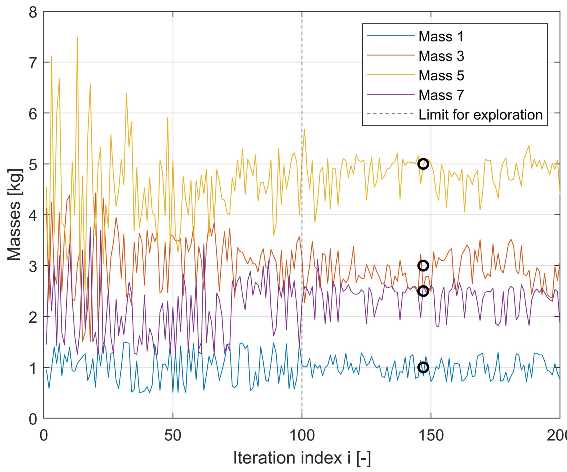
5.2. Results



(a)

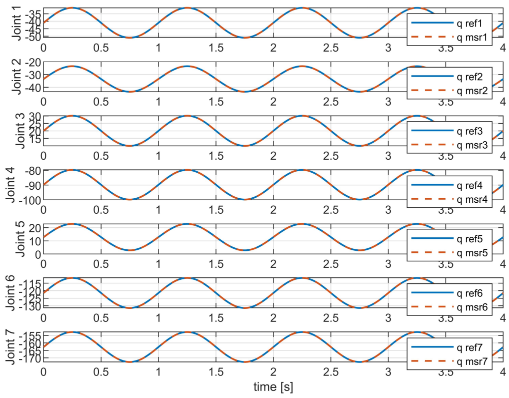


(b)

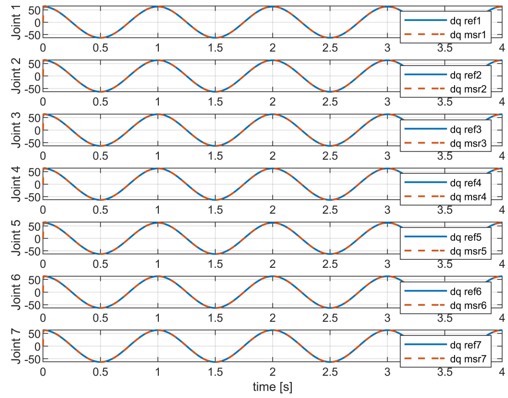


(c)

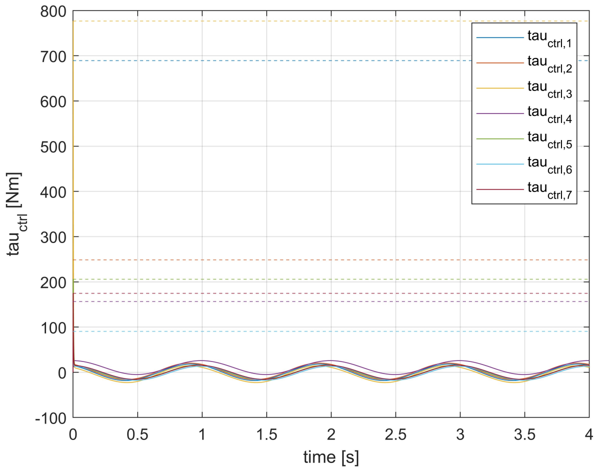
Figure 10: Results from mass estimation: a) acceleration profile compared against reference, b) joint torque profiles c) mass candidates evolution



(a)



(b)



(c)

Figure 11: Results from PID optimization: a) trajectory profile compared against reference, b) velocity profile compared against reference, c) joint torque profiles

As shown in table tab. 14, this optimization method proves effective not only for tuning the PID gains, but also for estimating the inertias of each joint.

Joint	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7
Real mass [kg]	1.0000	0	3.0000	0	5.0000	0	2.5000
Estimated mass [kg]	1.0031	/	2.9972	/	5.0056	/	2.4991

Table 14: Real and estimated mass values for each joint

As observed in fig. 11, the algorithm is again able to faithfully reproduce the assigned reference trajectories even in the presence of small variations in the mass values compared to the real ones. This is due to the effectiveness of the controller, which is either insensitive to such small variations or capable of compensating for these minor mass estimation errors.

KPI	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7	Mean
mean_pos_abs	0.000641	0.000225	0.000339	0.000587	0.000661	0.000384	0.000620	0.000494
max_pos	0.00120	0.00110	0.00096	0.00094	0.00110	0.00100	0.00110	0.00106
var_pos_abs	$1.03{\cdot}10^{-7}$	$1.46{\cdot}10^{-8}$	$8.69{\cdot}10^{-8}$	$1.06{\cdot}10^{-7}$	$1.00{\cdot}10^{-7}$	$9.66{\cdot}10^{-8}$	$9.57{\cdot}10^{-8}$	$8.82{\cdot}10^{-8}$
mean_vel_abs	0.0042	0.0052	0.0040	0.0036	0.0040	0.0036	0.0040	0.0041
max_vel	1.0966	1.0966	1.0966	1.0966	1.0966	1.0966	1.0966	1.0966
var_vel_abs	$2.43{\cdot}10^{-4}$	$4.07{\cdot}10^{-4}$	$3.90{\cdot}10^{-4}$	$4.13{\cdot}10^{-4}$	$4.13{\cdot}10^{-4}$	$4.11{\cdot}10^{-4}$	$4.11{\cdot}10^{-4}$	$3.84{\cdot}10^{-4}$

Table 15: Selected KPI metrics following masses estimation

	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7
K_p	5175.2	704.31	560.71	5469.8	8801.1	6987.1	5025.9
K_i	1043.9	225.44	448.53	65.805	573.95	884.22	893.43
K_d	454.6	486.25	519.45	533.47	472.98	495.18	477.31

Table 16: Optimal PID parameter set following masses estimation

6 | Conclusions

The conducted study explored multiple optimization strategies, evaluating both the design of cost functions and the architecture of the optimization approach.

6.1. Cost Functions Comparison

Two distinct cost functions were tested and compared: one defined in the joint space, and one in the workspace. Each function was crafted to emphasize different performance objectives:

- **Joint-space cost function:** prioritizes precise control over individual joint angles, making it well-suited for tasks requiring fine motion coordination at the actuator level
- **Workspace cost function:** focuses on minimizing the end-effector trajectory error in cartesian space, and is ideal for tasks such as reaching, manipulation, and contact-based operations

The experimental results revealed that the choice between these cost functions significantly affects the optimization outcome. For example, when precise path tracking of the end-effector is essential, the workspace-based formulation demonstrated better global performance. Conversely, when joint-level coordination was prioritized, the joint-space formulation offered more controllable behaviour.

As shown in fig. 12 and fig. 13, the joint-space cost function results in smaller errors in terms of joint positioning, whereas the workspace cost function yields better accuracy in terms of end-effector position. This is further confirmed by the 3D plot in fig. 14, which illustrates the end-effector trajectory tracking.

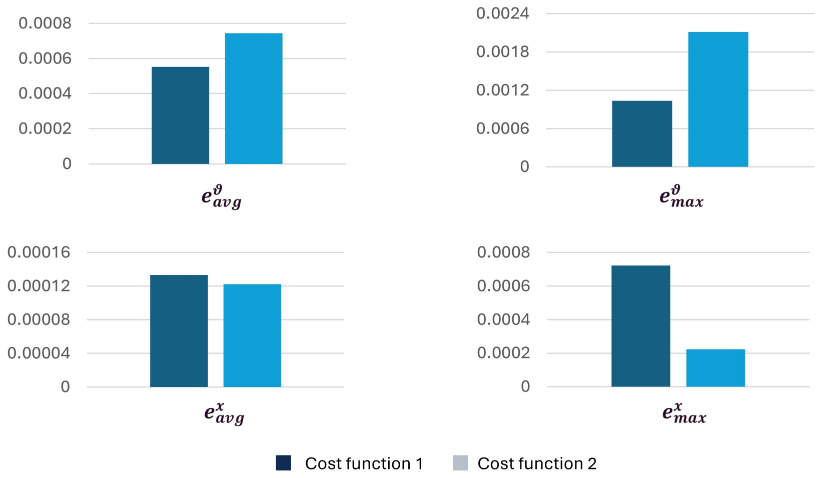


Figure 12: Comparison between two cost functions based on performance indicators: e^{θ}_{avg}) average tracking error in joint space, e^{θ}_{max}) maximum tracking error in joint space, e^{x}_{avg}) end-effector average tracking error in cartesian space, e^{x}_{max}) end-effector maximum tracking error in cartesian space

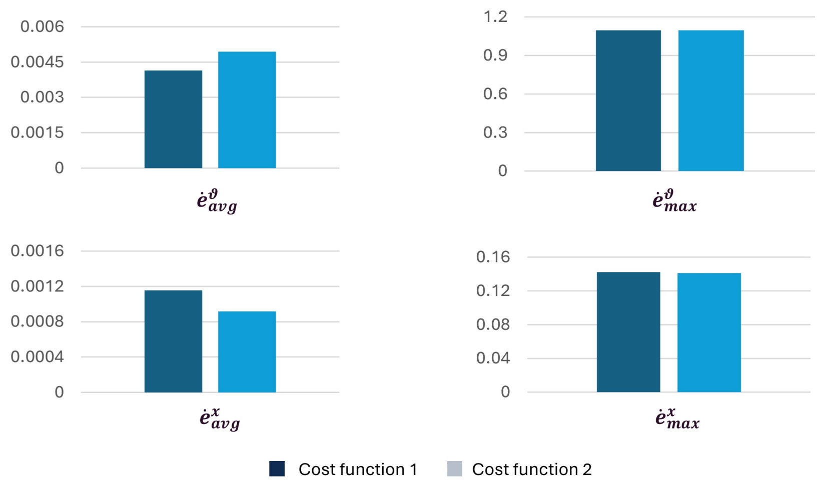
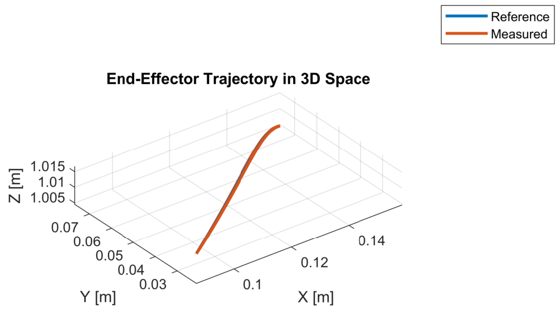
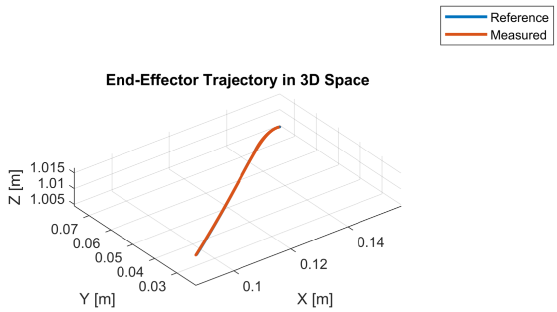


Figure 13: Comparison between two cost functions based on performance indicators: $\dot{e}^{\theta}_{avg}$) average velocity error in joint space, $\dot{e}^{\theta}_{max}$) maximum velocity error in joint space, $\dot{e}^{x}_{avg}$) end-effector average velocity error in cartesian space, $\dot{e}^{x}_{max}$) end-effector maximum velocity error in cartesian space



(a)



(b)

Figure 14: a) joint-space cost function, b) work-space cost function

6.2. Approaches Comparison: One-Shot vs Cascade

In addition to cost functions, two optimization architectures were tested:

- **One-shot optimization**: all 21 PID gains were optimized simultaneously in a high-dimensional space. Despite the computational cost, this approach allowed for a more effective overall tuning, as it accounted for the coupling effects among the joints and optimized their interaction as a whole.
- **Cascade optimization**: each joint was optimized independently via a separate progressive BO process (7 sequential runs). This approach reduced computational complexity due to dimensionality reduction, but neglected inter-joint dependencies, leading to suboptimal global performance.

This effect can be observed in fig. 15, where the cascade method results in slightly higher tracking errors.

Figure 15: Comparison between two approaches based on performance indicators: e^{θ}_{avg}) average tracking error in joint space, e^{θ}_{max}) maximum tracking error in joint space, $\dot{e}^{\theta}_{avg}$) average velocity error in joint space, $\dot{e}^{\theta}_{max}$) maximum velocity error in joint space

6.3. Summary of Trade-offs

The analysis highlights the trade-offs between global accuracy and problem dimensionality. The cascade approach, by optimizing each joint independently, benefits from a reduced search space and simpler sub-problems, making it a reasonable choice when a lighter optimization process is acceptable. However, it should be avoided when joint coupling or coordinated behavior is critical. In contrast, the one-shot approach considers all interactions simultaneously, offering more robust and globally coherent results at the expense of a higher computational burden.

In conclusion, the choice of cost function and optimization architecture must be tailored to the application-specific requirements. Workspace-based objectives are better suited for tasks focused on precise motion in cartesian space, while joint-based formulations offer greater control authority at actuator level. Similarly, one-shot optimization is preferable when global coherence is required, while cascade is acceptable under limited computational resources or when joint independence can be assumed.